

Software Integration & Maintenance

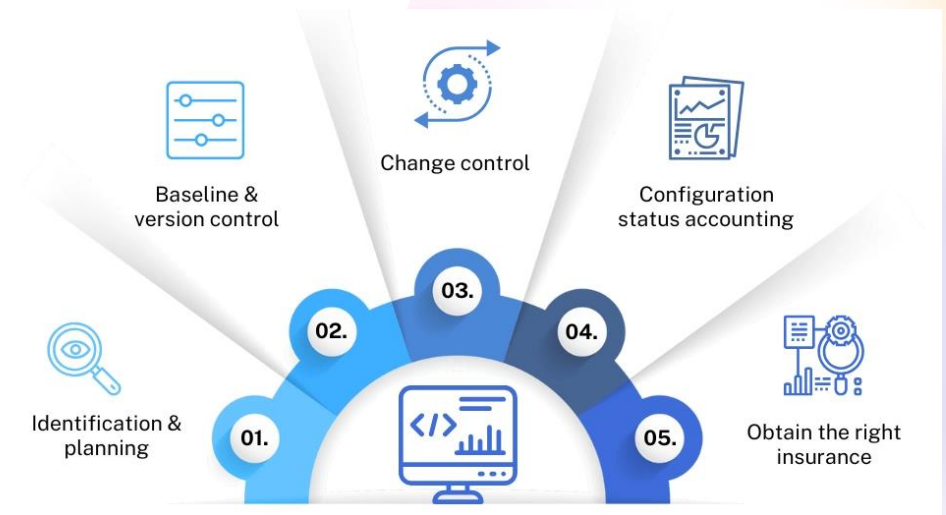
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Software Configuration Management (SCM)

- ❑ **SCM** is an **umbrella activity** in software engineering.
- ❑ It helps **manage and control changes** in software projects.
- ❑ It starts from the **beginning** of the project and continues till the **end**.
- ❑ **Everyone** involved in development uses **SCM**.



Why SCM is Important

- ❑ Software constantly changes due to **new requirements**.
- ❑ SCM prevents **confusion** and **mistakes**.
- ❑ It ensures the **correct version** of software is always used.
- ❑ It helps **teams work together** smoothly.

Real-Life Example of SCM

Imagine a group project report:

- ❑ One student edits the **introduction**.
- ❑ Another edits the **conclusion**.
- ❑ SCM is like Google Drive version history — it tracks who changed what and when.

Who is Involved in SCM?

- ☐ Developers
- ☐ Testers
- ☐ Project managers
- ☐ Customers and stakeholders
- ☐ A designated SCM manager or administrator

Sources of Change

- ☐ New business or market conditions
- ☐ New customer needs
- ☐ Business growth or downsizing
- ☐ Budget or schedule constraints

SCM Tools

Some popular SCM and automation tools:

- ❑ **SaltStack**
- ❑ **SmartFrog**
- ❑ **Puppet**
- ❑ **Git** (commonly used version control system)



Main Tasks of SCM

❑ Configuration Identification

- Identifies all items that need to be controlled.
- These are called Configuration Items (CIs).

Examples: Source code, design documents, test plans, user manuals.

In a web project:

1. index.html
2. style.css
3. login.js
4. Database schema
5. User manual

Main Tasks of SCM

❑ Version Control

- Keeps track of different versions of software.
- Allows developers to go back to older versions if needed.
- Prevents overwriting others' work.

Examples:

You save your project as: report_v1.docx,
report_v2.docx, report_final.docx.

Version control systems like Git automate this process.

Main Tasks of SCM

❑ Semantic Versioning (SemVer)

Version format: MAJOR.MINOR.PATCH

Example: 1.8.5

- MAJOR: Big changes
- MINOR: New features
- PATCH: Bug fixes

Major version change: 1.0.0 → 2.0.0 (breaking changes)

Minor version change: 1.2.0 → 1.3.0 (new features)

Patch version change: 1.2.1 → 1.2.2 (bug fixes)

Main Tasks of SCM

❑ Semantic Versioning (SemVer)

Rules of Semantic Versioning:

- Each number must increase.
- You cannot change a released version.
- Patch resets when minor increases.
- Patch and minor reset when major increases.

Main Tasks of SCM

❑ Semantic Versioning (SemVer)

Pre-release Versions:

- Used to test software before final release.

Example: 1.0.0-alpha, 1.0.0-beta, 1.0.0-rc.1

Final release: 1.0.0

Main Tasks of SCM

❑ Change Control

- A systematic process to manage changes.
- Ensures only necessary and approved changes are made.
- Prevents unexpected problems in the system.

Why Change Control is Needed

- Avoids unnecessary changes
- Documents all changes
- Prevents service disruption
- Uses resources efficiently

Main Tasks of SCM

❑ Engineering Change Order (ECO)

A formal document that requests a change.

➤ Includes:

- Description of change
- Reason for change
- Impact analysis
- Approval status

Main Tasks of SCM

❑ Configuration Auditing

- Ensures configuration items are correct and complete.
- Checks whether changes were properly implemented.
- Confirms system matches documentation.

Example:

- Checking if the delivered software matches the design document.
- Verifying all approved changes were correctly applied.

Main Tasks of SCM

❑ Status Reporting

Provides current information about:

- Versions
- Changes
- Configuration items

Helps managers, developers, and customers stay informed.

Documents Generated in SCM

- ☐ Admin guides
- ☐ User manuals
- ☐ FAQs
- ☐ Release notes
- ☐ Installation guides

Benefits of SCM

- ❑ Better control over software changes
- ❑ Reduced errors and confusion
- ❑ Improved teamwork
- ❑ Higher software quality
- ❑ Easier maintenance

Thank You.